A Cascade Rotated Anchor-Aided Detector for Ship Detection in Remote Sensing Images

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***Abstract*— Automatic ship detection in high-resolution remote sensing images has attracted increasing research interest due to its numerous practical applications. However, there still exist challenges when directly applying state-of-the-art object detection methods to real ship detection, which greatly limits the detection accuracy. In this article, we propose a novel cascade rotated anchor-aided detection network to achieve high-precision performance for detecting arbitrary-oriented ships. First, we develop a data preprocessing embedded cascade structure to reduce large amounts of false positives generated on blank areas in large-size remote sensing images. Second, to achieve accurate arbitrary-oriented ship detection, we design a rotated anchor-aided detection module. This detection module adopts a coarse-to-fine architecture with a cascade refinement module (CRM) to refine the rotated boxes progressively. Meanwhile, it utilizes an anchor-aided strategy similar to anchor-free, thus breaking through the bottlenecks of anchor-based methods and leading to a more flexible detection manner. Besides, a rotated align convolution layer is introduced in CRM to extract features from rotated regions accurately. Experimental results on the challenging DOTA and HRSC2016 data sets show that the proposed method achieves 84.12% and 90.79% AP, respectively, outperforming other state-of-the-art methods.**

***Index Terms*— Cascade rotated anchor-aided detector, remote sensing images, ship detection.**

I. Introduction

THE development of optical remote sensing imaging technology greatly promotes the studies in geospatial object detection, such as ships [1], airplanes [2], and vehicles [3]. Among these research topics, ship detection plays an important role in its numerous practical applications, such as environmental protection, harbor monitoring, and marine rescuing. Automatic ship detection aims to localize and classify the ships in satellite images simultaneously. Large amounts of easily accessible high-resolution images provide more detailed information of ships to promote the detection accuracy. Thus, ship detection in remote sensing images has attracted increasing research interest in recent years [4]–[7].

Most of the traditional ship detection methods utilize statistical distributions or handcrafted features to distinguish ships from backgrounds. However, this type of feature mainly depends on low-level information, which is greatly affected by the background. This may bring some drawbacks, such as poor feature representation ability and low generalization capacity to images with complex backgrounds. On the other hand, with the rapid development of deep convolutional neural networks (CNNs), CNN-based methods have largely promoted the performance of object detection in natural images [8]–[14]. Generally, state-of-the-art object detection methods can be divided into two categories, i.e., anchor-based detectors and anchor-free detectors. The former has to predefine a set of anchors with various sizes and aspect ratios and then predict the object category and location. Representative detectors include RetinaNet [13], single-shot multibox detector (SSD) [11], Faster R-CNN [10], and Mask R-CNN [15]. In contrast, anchor-free detectors, such as CornerNet [16], CenterNet [17], and FCOS [18], eliminate the process of complicated anchor design and can achieve comparable performance with anchor-based ones. Encouraged by the above object detection methods in natural images, some CNN-based geospatial object detection methods have been proposed to pursue better detection performance in remote sensing images [19]–[22].

However, due to huge differences in way of the image taken (e.g., distance and orientations) between natural images and remote sensing images, there still exist challenges when directly applying state-of-the-art object detection methods to ship detection, greatly limiting the detection performance.

1) High-resolution remote sensing images usually have a big size and require large memory. To mitigate this problem, the images are often cropped into small patches first and then detected in a patchwise manner [see Fig. 1(a)]. However, the targets in images usually only occupy a small area, resulting in large amounts of blank patches without targets. Therefore, detecting these blank patches will inevitably lead to numerous false positives, thus reducing the detection accuracy.
2) As illustrated in Fig. 1(b), objects in natural scenarios are generally horizontal, while ships in remote sensing images are often arranged arbitrary-oriented. Compared with a horizontal bounding box, a rotated bounding box is more appropriate for representing objects with arbitrary orientations, which further increases the difficulty of accurate detection.

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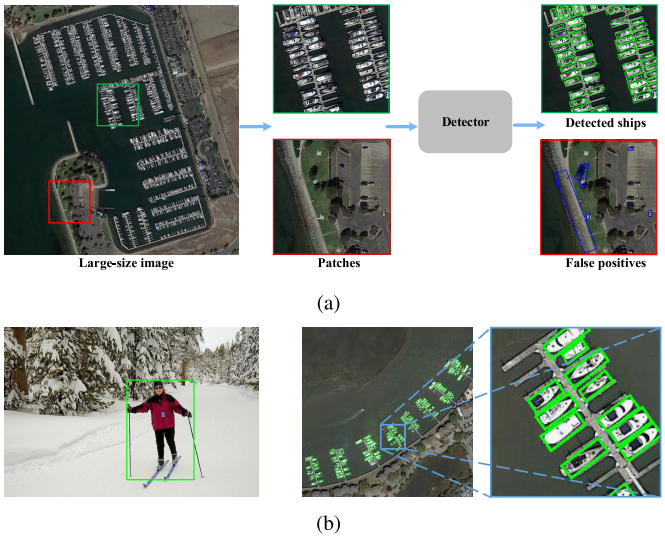


Fig. 1. (a) Workflow of detection on large-size images. (b) Illustration of horizontal object detection in natural images and rotated ship detection in high-resolution remote sensing images.

3) Previous ship detection methods are mainly anchor-based, which depends heavily on the anchor setting. As the performance is greatly affected by the anchor-related hyperparameters, these methods need to adjust the anchors carefully for a specific data set, leading to poor generalization ability.

To solve the aforementioned problems, this article proposes a novel **cascade rotated anchor-aided network** (CR2A-Net) for ship detection in high-resolution remote sensing images. This network aims to achieve high-precision performance for rotated ship detection in a more flexible manner.

First, to deal with the blank patches caused by cropping operation for high-resolution images, we develop a cascade structure by inserting a data preprocessing module before the rotated anchor-aided detection module. The former is employed to determine the existence of ships in the input image, and the latter is followed to detect the ships accurately if available. The existence of a data preprocessing module enables the network to filter the blank patches without targets before detecting, which is beneficial to improve the detection performance.

Second, to address the difficulty of detecting arbitrary-oriented ships, we present an accurate and robust coarse-to-fine detector in the stage of rotated anchor-aided detection, achieving high detection performance. The detector consists of a basic detector (BD) and a cascade refinement module (CRM). The former is used to obtain a preliminary rotated box at the position where ships may exist, and the latter further modifies the rotated box to achieve more accurate detection results. The proposed coarse-to-fine detector is innovative due to two points. 1) the basic one-stage detector in our model adopts a new anchor-aided strategy, leading to fewer anchors and a more flexible structure and 2) we add a CRM to the BD, which is specially designed for one-stage rotated object detection. Due to this well-designed CRM, the detector can refine the rotated box more accurately, which enables the network to obtain high detection accuracy. To be specific, a critical issue when cascading one-stage methods is feature misalignment between the extracted features and the regressed box. Therefore, we propose a novel rotated align convolution (RAC) at the front of the CRM to align the receptive fields of the convolution kernel and the rotated box of detected ships. Through this process, the corresponding features of rotated regions can be extracted more effectively, which has a significant impact on the detection performance.

In this article, the proposed method aims to extract more robust rotation-invariant features to promote location accuracy, which is consistent with previous methods for rotated object detection. However, they are different in some aspects. First, different from these methods that manually design a set of rotated anchors [5] or generate rotated anchors via a complex subnetwork [23], we simplify the process by setting only one anchor per location with a fixed size regardless of data sets. This improves the robustness of the network when applying to different data sets with different object sizes and orientations. Second, we attach a CRM to refine the rotated box. Unlike previous methods, such as Cascade R-CNN [14] that improves on two-stage method Faster R-CNN [10] by utilizing multiple R-CNN stages, the CRM in CR2A-Net is attached to a one-stage detector. On the one hand, due to the removal of the region proposal network (RPN) stage that requires nonmaximum suppression (NMS) operation and proposal selection, our detector keeps a more simplified fully convolutional structure and is easier to train. On the other hand, the proposed RAC layer in CRM replaces the complex rotated region of interest (RoI) Align operation in R-CNN structure, which explicitly extracts rotation-invariance features from corresponding rotated ship areas accurately with fewer parameters. Moreover, different from deformable convolution whose offsets are learned during training, the RAC can obtain more precise and effective rotation-invariance features.

In summary, the contributions of this work can be concluded as follows.

1) We develop an efficient and effective cascade structure embedded with a data preprocessing module (DPM). This design can not only improve detection efficiency but also reduce large amounts of false positives, thus improving detection performance.
2) We design an accurate and robust coarse-to-fine detector for arbitrary-oriented ships. The anchor-aided strategy is employed to avoid complicated anchor design and, thus, improves the robustness of the detector. Moreover, we propose a novel RAC in the detector to extract rotation-invariance features from corresponding rotated regions sufficiently.
3) Without bells and whistles, our method outperforms state-of-the-art methods on the challenging DOTA[1] [24] and HRSC2016[2] [25] data set and achieves a detection accuracy up to 84.12% and 90.79%, respectively.

II. Related Work

A. General Object Detection

The recent rapid development of deep neural networks has greatly improved the performance of object detection. Current

[1]https://captain-whu.github.io/DOTA/dataset.html
[2]https://www.kaggle.com/guofeng/hrsc2016

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object detection technology can be divided into two categories: anchor-based detectors and anchor-free detectors.

1) Anchor-Based Detectors: Anchor-based detectors can be roughly divided into two categories: two-stage methods and one-stage methods. The classical two-stage method Faster R-CNN [10] contains an RPN stage for generating candidate proposals and a region-based prediction network (R-CNN) for detecting targets [8], [9]. Subsequently, a series of methods have been proposed [12], [14], [26], [27] to improve on this framework. Two-stage methods can usually achieve better detection performance while having a slow inference speed. Compared with the two-stage methods, one-stage anchor-based methods have a simplified structure and can achieve a faster inference speed. The classic one-stage method SSD [11] removes the RPN stage and directly predicts the classification score and bounding box regression offsets. RetinaNet [13] pointed out that a key problem restricting the performance of the one-stage method is the imbalance between positive and negative samples and proposes focal loss to alleviate this problem. Other than these horizontal detection methods, numerous rotated object detection methods have also been proposed [28], [29]. Some early works adopt handcrafted rotation-invariance features to represent the rotated objects, thus limiting the detection performance. Wu *et al.* [30] learned rotation-invariance features based on the spatial-frequency channel feature that considers the channel features in spatial and frequency domains jointly. Wu *et al.* [31] extracted the feature representations by refining the Fourier-based rotation-invariant feature with aggregate channel features. In contrast, CNN-based methods have shown more powerful feature extraction abilities and achieved favorable performance. Cheng *et al.* [32] proposed a novel training strategy to achieve rotation-invariance, which applies rotation data augmentation to training samples first and enforces the feature representations of the training samples before and after rotating to be mapped close to each other. Li *et al.* [33] addressed the rotation problem by introducing a rotation-insensitive RPN that includes additional anchors with orientations besides the conventional horizontal anchors. Based on Faster R-CNN, R2CNN [28] added a new branch to detect rotated bounding box with two points and the heights of corresponding boxes. RRPN [29] adopted anchors with multiple orientations in the RPN stage and proposes the rotated RoI (RRoI) pooling layer in the R-CNN stage to extract the features of the rotated proposals. Although these anchor-based methods have achieved outstanding performance, they require a lot of effort to adjust the hyperparameters related to anchor setting.

2) Anchor-Free Detectors: Anchor-free detectors can be roughly classified into two categories: keypoint-based methods and pixel-based methods. Keypoint-based methods first detect the keypoints and then locate the bounding box according to these keypoints. CornerNet [16] detected the top-left and bottom-right corner points to locate the bounding box. ExtremeNet [34] detected four vertices and a center point to locate the bounding box. CenterNet [17] detected the center point and predicts other attributes of the object such as size, 3-D position, orientation, and posture. In contrast, pixel-based anchor-free methods predict a classification score and a bounding box at each pixel point directly. FCOS [18] locates the object by predicting the distance from the pixel to the four sides of the bounding box. FoveaBox [35] directly predicts the coordinates of the four corner points of the bounding box. There are also some anchor-free methods for rotated object detection. EAST [36] proposed a novel method for text detection, which predicts the distance of each point to the four sides of the oriented bounding box and angle information at each pixel. Lyu *et al.* [37] generated a corner map and a position-sensitive segmentation map to calculate the location and classification score of each bounding box, respectively. In addition, some methods applying the idea of anchor-free to two-stage methods have been proposed. Wang *et al.* [19] proposed an anchor-free deconvolutional network with the region generation module to replace the original RPN and abandon the anchor mechanism. Yu *et al.* [23] proposed an anchor-free orientation guided anchor generation subnetwork to produce rotated anchors automatically instead of handcrafted design. These anchor-free methods do not need to manually predefine anchors, which simplifies the object detection framework greatly and can achieve comparable performance as well.

B. Ship Detection

Ship detection in remote sensing images has been studied for many years. Most of the early works [38]–[42] employ handcrafted features to distinguish between ships and backgrounds, resulting in very limited detection performance. In the past few years, inspired by the general object detection methods in natural images, many CNN-based methods have been proposed in remote sensing images, especially for detecting ships or other objects. Kang *et al.* [22] combined Faster R-CNN with traditional method and regard the region proposal generated by RPN as a guide window for the constant false-alarm rate (CFAR) algorithm. Wang *et al.* [20] proposed a one-stage detection network based on RetinaNet to detect ships in synthetic aperture radar (SAR) images. Li *et al.* [1] proposed an end-to-end network based on Faster R-CNN and design a hierarchical selective filtering layer to generate better features for detecting multiscale ships. As ships in remote sensing images are typically distributed in arbitrary directions, some rotated object detection methods are proposed to accurately locate arbitrary-oriented ships. The work in [4] is based on the one-stage network YOLOv2 [43] and takes the angle information into consideration. Yang *et al.* [5] added a branch to Faster R-CNN to regress the rotated bounding boxes. Guo *et al.* [6] proposed a rotational Libra R-CNN to achieve rotated ship detection and address the imbalance problems during training. Li *et al.* [7] proposed a more effective feature extraction method, which extracts the multilevel and multiscale features via a spatially variant adaptive pooling layer and a shape adaptive pooling layer, respectively. Xia *et al.* [24] constructed a large-scale data set for object detection in remote sensing images and build a rotational network to better detect various objects, including ships. RoI-Transformer [44] first transforms the horizontal proposals to rotated proposals using RRoI learner and then extracts rotation-invariant features from rotated regions via a Rotated Position Sensitive RoI

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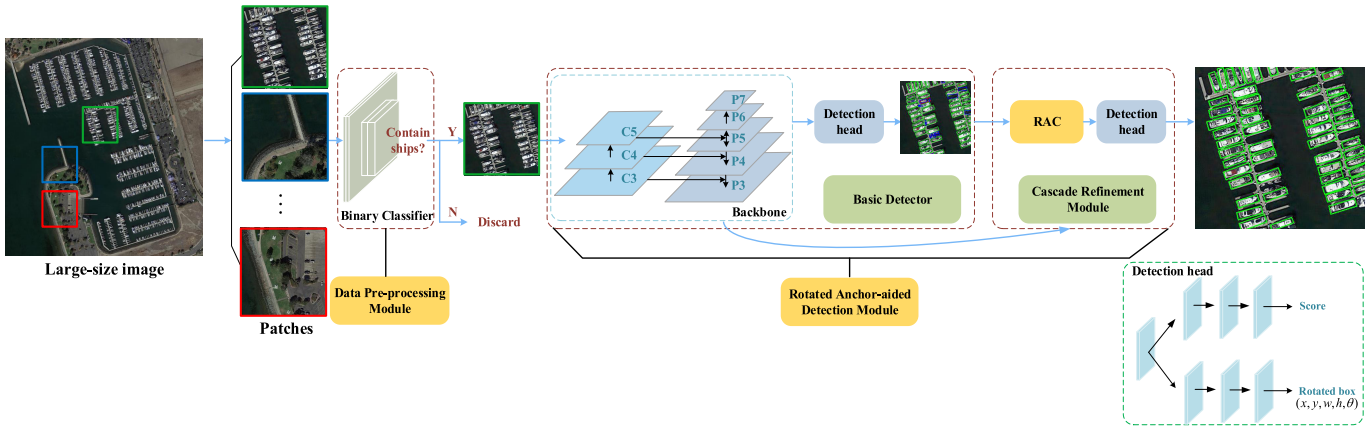


Fig. 2. Overview of the proposed CR2A-Net. A large-size input image is first divided into patches and then fed to a DPM to judge if it contains ships or not. The one containing targets will go through the subsequent rotated anchor-aided detection module; otherwise, it will be discarded. The BD produces a preliminary result, and the CRM refines the rotated boxes to get the final detection results. RAC is employed to extract features from rotated regions accurately.

Align module. CAD-Net [45] employs the global and local contextual information of the image to guide the network to focus on more informative regions.

Different from the abovementioned anchor-based detection methods, the proposed cascade rotated anchor-aided detection network for arbitrary-oriented ship detection adopts a novel anchor-aided strategy inspired by anchor-free methods. As there is no need to manually design the anchors, the proposed method is more flexible and computationally efficient than previous methods.

C. Cascade Object Detectors

Several studies have shown that cascade multiple stages can promote object detection performance [14], [46], [47]. Cascade R-CNN [14] adopts multiple R-CNN stages to perform classifications and regressions multiple times, significantly improving the classification and localization accuracy. Similarly, Cascade RPN [46] employs multiple RPN stages to obtain high-quality proposals. However, when applying the same idea to one-stage detectors, the feature misalignment between the RoI and the extracted features needs to be addressed. The misalignment is caused because the center point of refined anchors is beyond the pixel point. Unlike the cascade two-stage methods that use RoI Pooling or RoI Align to extract features, cascade one-stage methods should maintain a fully convolutional structure to keep the speed advantage. Cascade RetinaNet [47] employs deformable convolution to attain feature alignment. However, its offset parameters are learned automatically from the feature map using additional convolutional layers, which cannot ensure that the features are accurately aligned. AlignDet [48] also employs deformable convolution, but its offsets are obtained by calculating the offsets between the refined anchors and the predefined anchors. Thus, it can extract features exactly from the refined regions.

We propose a coarse-to-fine detector with a CRM to achieve high-precision performance for detecting rotated ships. Different from the previous methods designed for horizontal object detection, we propose a RAC layer to address the feature misalignment in CRM, which is specially designed for rotated object detection.

III. Proposed Method

Fig. 2 illustrates the framework of the proposed CR2A-Net for arbitrary-oriented ship detection in remote sensing images. Our CR2A-Net consists of two parts: 1) a DPM that distinguishes the patches including ships from those blank ones and 2) a rotated anchor-aided detection module that predicts rotated bounding boxes accurately in a flexible manner. The detection network adopts a coarse-to-fine architecture, which is composed of a BD and a CRM. BD is built on the structure of classical one-stage detection network RetinaNet [13], which produces a preliminary detection result. CRM is attached to BD to refine the rotated bounding box progressively. Moreover, RAC is proposed in CRM to extract features accurately from rotated regions.

A. Data Preprocessing Module

Processing large-scale images in remote sensing fields directly requires high memory. A common solution is to crop the large images into small patches first and then merge the detection results of these patches to obtain the final detections of the original images. However, an image contains only a few targets in most cases, which occupies only a small area. Therefore, the small patches obtained by cropping operation may not contain the targets [see Fig. 1(a)]. Detecting these empty patches will not only reduce the detection efficiency but also cause many false positives. Therefore, we propose a cascade structure by adding a DPM in front of the detector. The DPM is used to discriminate if an image contains ships or not before detecting on the image. This cascade structure can significantly reduce false positives generated on the empty patches, thus improving the detection performance greatly.

In practice, we use a binary classifier to implement the DPM. The binary classifier adopts ResNet34 [49] pretrained on ImageNet [50]. The architecture is shown in Table I. It consists of a downsampling step by a convolution and pooling operation followed by four layers of similar structure. Each of the layers contains several residual blocks, and each residual block is composed of two layers of 3×3 convolution with a fixed number of filters, i.e., 64, 128, 256, and 512, respectively. Besides, a shortcut connection that achieves identity mapping

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TABLE I
ARCHITECTURE OF RESNET34 USED IN OUR DPM

layer_name	architechture
conv_1	conv, 7×7, 64, stride 2 max pool, 3×3, stride 2
conv_2	$\begin{bmatrix} conv, 3 \times 3, 64 \\ conv, 3 \times 3, 64 \end{bmatrix} \times 3$
conv_3	$\begin{bmatrix} conv, 3 \times 3, 128 \\ conv, 3 \times 3, 128 \end{bmatrix} \times 4$
conv_4	$\begin{bmatrix} conv, 3 \times 3, 256 \\ conv, 3 \times 3, 256 \end{bmatrix} \times 6$
conv_5	$\begin{bmatrix} conv, 3 \times 3, 512 \\ conv, 3 \times 3, 512 \end{bmatrix} \times 3$
	global average pooling 1000-d fc, softmax

is inserted to address the degradation problem of network optimization during training. Batch normalization and rectified linear unit (ReLU) activation are adopted right after every convolution layer. The network ends with a global average pooling layer and a 1000-way fully connected layer with softmax activation. The last layer of the network is replaced with a fully connected layer with one neuron. Note that the backbone here can also be replaced with other popular CNNs, and we adopt ResNet34 to balance the computational efficiency and accuracy. The cross-entropy loss is used for binary classification, i.e.,

$$\mathrm{CE}(p, q) = \begin{cases} -\log(p), & \text{if } q = 1 \\ -\log(1-p), & \text{otherwise} \end{cases} \tag{1}$$

where $q \in \{0, 1\}$ is the ground-truth class label, and 1 denotes that the image contains ships, while 0 means that it is empty. $p \in [0, 1]$ is the model's estimated probability of class label $q = 1$.

One crucial problem to note is the imbalanced positive and negative samples, which is not conducive to stable training of the network. For example, the ratio of empty and nonempty images after cropping operation in the DOTA training set is about 12 : 1. To alleviate this imbalance problem, we adopt the methods in [51]. Especially, we employ the bias initialization method during training to set the bias of last fully connected layer to $b = -\log(N/N_c - 1)$, where N is the total number of samples and N_c is the number of positive samples that contain ships. This strategy can reduce the large loss caused by the dominant-negative class in early training. During inference, we adopt the confidence shifting method to change the threshold of classification from 0.5 to $\theta_T = (N_c/N)$, i.e., if $p > \theta_T$, the image is classified as containing ships; otherwise, it is classified to be empty. In this way, the network can focus more on the ship class and avoid ignoring some images containing targets.

In the inference stage, images are cropped into small patches first and then delivered to the classifier in a patchwise manner. Those images classified as containing targets will be sent to the subsequent detector to generate detection results, while the empty images are filtered and regarded as not including any detection target. Finally, all the detection results of patches are integrated to yield the final detections of original large-size images. Helped by the classifier, about 80% of the patches do not need to go through the subsequent burdensome detection process, which improves the detection efficiency and greatly reduces false positives brought by empty patches.

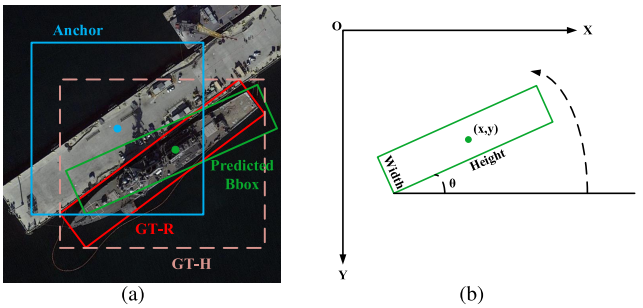


Fig. 3. (a) Illustration of different boxes on an input image. GT-R denotes the GT-R box and GT-H denotes the GT-H box. (b) Definition of rotated object representation.

B. Rotated Anchor-Aided Detection Module

General object detection in natural images is horizontal detection, i.e., the bounding box is a horizontal rectangle represented as (x, y, w, h), where x, y, w, and h denote the box's center point, width, and height, respectively. This representation is not suitable for ship detection in remote sensing images where the objects are often arbitrarily oriented. Therefore, rotated object representation and detectors are proposed to detect the objects with arbitrary orientations. Especially, a rotated bounding box is represented as (x, y, w, h, θ), where the extra variable θ denotes the angle information of the rotated box. As shown in Fig. 3, θ is defined as the angle between the long side of rotated box and the x-axis, and $\theta \in (-180°, 0°)$. Our rotated anchor-aided detection module adopts a coarse-to-fine structure that is specially designed for accurate rotated ship detection, and it can be divided into a BD that predicts a preliminary rotated box in a flexible manner and a CRM to further refine the rotated bounding box.

1) Basic Detector: Most of the previous rotated object detection methods are anchor-based that tiles multiple anchors with different sizes and aspect ratios at each position. Therefore, these methods have some drawbacks inherited from other anchor-based methods. First, some anchor-related hyperparameters, such as the size, aspect ratio, and number of anchors, need to be adjusted manually, which requires a lot of effort. Second, these hyperparameters depend heavily on specific data sets, so it is not conducive to the generalization ability when transferred to another data set with different object sizes or aspect ratios. Third, placing multiple anchors at each location will reduce the computational efficiency and further expand the imbalance between positive and negative samples during training. Recently, several anchor-free methods have been proposed in general object detection fields to tackle the above problems. Inspired by these methods, we propose an anchor-aided detector similar to anchor-free methods for rotated ship detection, which can predict rotated bounding boxes accurately. Due to the removal of anchor-related restrictions, our method is more flexible and computationally efficient.

The architecture of BD is shown in Fig. 2. The network can be divided into a backbone network and a detection head. The backbone network adopts the FPN [12] architecture on the

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basis of ResNet [49] to extract multiscale features. ResNet is composed of several residual blocks, which contains a series of stacked convolutional layers and a shortcut connection. As the depth of networks increases, the optimization of the network becomes more difficult because of the vanishing/exploding gradients. The shortcut connection in residual block adds the shallow layer features via an identity mapping, which can alleviate the optimization problems and extract more effective feature representations for objects. Moreover, since the ship targets in data sets contain multiple different scales, we adopt a feature pyramid network to detect different sizes of objects on different levels of feature maps, which enables the model to obtain more effective semantic information. Especially, following FPN, we use the output of corresponding residual block of ResNet conv3, conv4, and conv5 as feature maps $C3$, $C4$, and $C5$. Then, we compute the multilevel features $\{P3, P4, P5\}$ (corresponding to $C3$, $C4$, and $C5$) through top–down and lateral connections (see Fig. 2). $P6$ is obtained by applying a 3×3 convolution with stride 2 on $C5$ for downsampling. $P7$ is computed by utilizing a 3×3 convolution with stride 2 on $P6$. Finally, we obtain multilevel features $\{P3, P4, P5, P6, P7\}$. The detection head contains a classification subnet and a regression subnet to predict the classification score and coordinates of the rotated box, respectively. Meanwhile, we set only one anchor at each position with a fixed size ranges from 32^2 to 512^2 on pyramid levels $P3$–$P7$, respectively. Compared with the original anchor-based RetinaNet that tiles a set of manually defined anchors of different sizes and aspect ratios at each location, our network has no need to consider anchor-related hyperparameter settings.

During the training stage, we need to define positive and negative samples. A common practice is to assign the labels according to the intersection-over-union (IoU) between anchors and ground-truth objects, which needs to manually set the threshold. Especially, the anchors with IoU greater than a higher threshold with any ground-truth box are selected as positive samples, the anchors with IoU less than a lower threshold for all ground-truth boxes are regarded as negative samples, and the rest are ignored. In the case of our anchor-aided strategy, only one anchor with a fixed size is set at each location, resulting in less matched positive samples than the original RetinaNet with a more complex anchor setting. Therefore, we adopt a novel adaptive training sample selection (ATSS) method proposed in [52] to define the positive and negative samples. Using this sampling method, each ground-truth can match more anchors, and the quantities of matched anchors between objects with different scales and aspect ratios are relatively balanced, improving the recall rate of small targets.

Considering the computational efficiency, the matching is between horizontal anchors and axis-aligned bounding rectangle of rotated ships, i.e., the IoU is calculated using horizontal anchors and ground-truth horizontal (GT-H) box [see Fig. 3]. The whole ATSS process can be divided into three steps. First, for each GT-H box g, we select n anchor boxes at each feature level whose centers are closest to the center of g in terms of the L2 distance. Assuming that there are K feature pyramid levels,

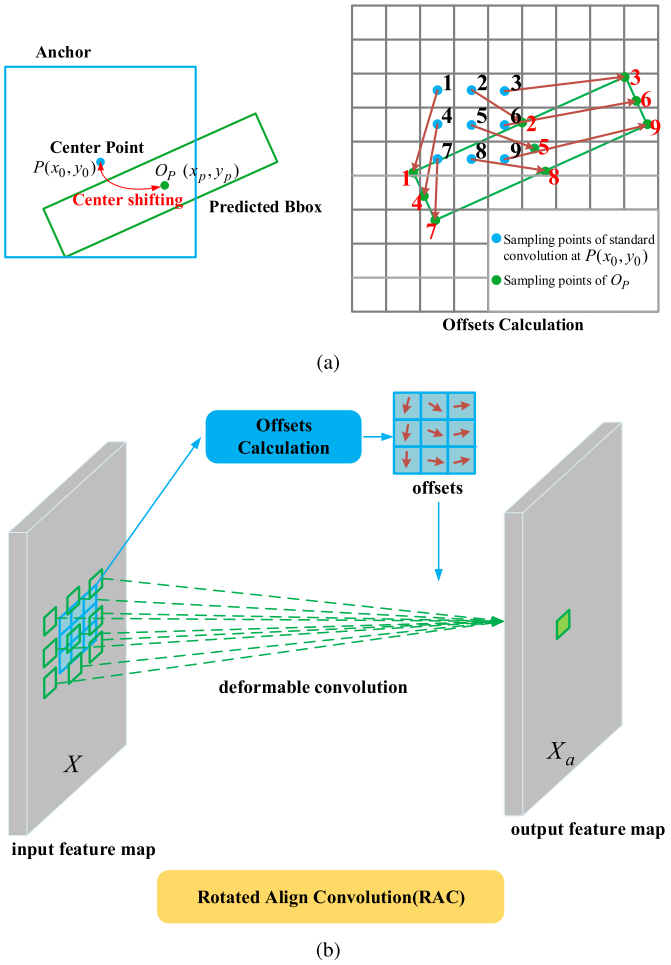


Fig. 4. (a) Left is the visualization of misalignment between anchor and predicted box, right is the offsets calculation between sampling points of standard convolution and rotated box. (b) Illustration of the proposed rotated align convolution.

each g will obtain $n \times K$ candidate positive samples in total. Second, we compute the IoU between these candidates and GT-H box g. m_g and v_g denote mean and standard deviation of the IoU, respectively. Finally, these candidates whose center is within the GT-H box and IoU is greater than or equal to the threshold $t_g = m_g + v_g$ are selected as the final positive samples. All remaining anchors except the positive samples are regarded as negative samples. In addition, if the anchor box is assigned to multiple ground-truth boxes, the box with the highest IoU will be selected.

The regression targets are calculated using the matched ground-truth rotated (GT-R) box. As mentioned earlier, a rotated bounding box is represented as (x, y, w, h, θ). For the horizontal anchors, we set θ to $-180°$ by default. Then, the regression targets can be formulated as

$$\begin{aligned} t_x &= (x_g - x_a)/w_a, t_y = (y_g - y_a)/h_a \\ t_w &= \log(w_g/w_a), t_h = \log(h_g/h_a), t_\theta = \theta_g - \theta_a \end{aligned} \tag{2}$$

$$\begin{aligned} t_x^* &= (x_p - x_a)/w_a, t_y^* = (y_p - y_a)/h_a \\ t_w^* &= \log(w_p/w_a), t_h^* = \log(h_p/h_a), t_\theta^* = \theta_p - \theta_a \end{aligned} \tag{3}$$

where x_g, x_a, and x_p denote the coordinates of the ground-truth box, anchor box, and predicted box, respectively (likewise,

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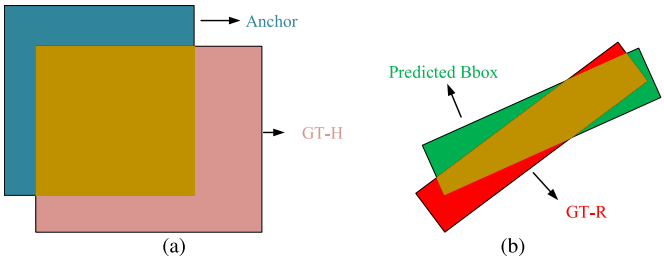


Fig. 5. Illustration of IoU between horizontal boxes and rotated boxes. (a) IoU between an anchor and a ground-truth horizontal bounding box (GT-H). (b) IoU between a predicted box and a ground-truth rotated bounding box (GT-R). The brown regions indicate the intersection area between two boxes.

for y, w, h, and θ). t_x^* represents the network output value corresponding to the predicted rotated box (likewise, for $t_x^*, t_y^*, t_w^*, t_h^*$, and t_θ^*). The multitask loss used for training is defined as

$$L = \frac{\lambda_1}{N_{\text{cls}}} \sum_i L_{\text{cls}}(p_i, l_i) + \frac{\lambda_2}{N_{\text{reg}}} \sum_i l_i \cdot L_{\text{reg}}\left(t_i^*, t_i\right) \quad (4)$$

where N_{cls} and N_{reg} represent the number of total anchors and positive anchors in a minibatch, respectively. i is the index of anchors in a minibatch. l_i is a binary class label, where $l_i = 1$ for positive anchors and $l_i = 0$ for negative anchors. p_i is the predicted probability of the corresponding class. t_i and t_i^* represent the target vectors of ground-truth and the predicted offset vectors, respectively. λ_1 and λ_2 are the hyperparameters that control the tradeoff of each loss term and are set to 1 in our experiments. L_{cls} and L_{reg} represent the classification loss and regression loss, respectively. We employ focal loss [13] and smooth $L1$ loss [9] for L_{cls} and L_{reg}, respectively.

2) Cascade Refinement Module for Rotated Object Detection: Although our one-stage detector BD achieves good performance, the regression is not accurate enough due to the challenge of regressing rotated angle of objects in rotated object detection. A common practice is to add multiple stages to refine the detection results iteratively, which can obtain more accurate confidence scores and higher localization performance. Therefore, we propose a CRM specially designed for our rotated ship detection task. When cascading stages in rotated object detection, a crucial problem to solve is to accurately extract the features from rotated regions. We address this problem by introducing a new RAC, and this operation turns out to be effective in our experiments.

As shown in Fig. 4(a), for each location, we can obtain an anchor box centered at this location and a rotated box predicted by BD. However, the center point of the corresponding rotated box predicted by BD is shifted compared with the original anchor. Using a standard convolution operation at this position may lead to a misalignment between the obtained features and the corresponding rotated box. To address this problem, we propose an RAC to extract the features from rotated regions accurately. The detailed process is as follows.

Given an input feature map X and a specific position $P(x_0, y_0)$, the features at P on the output feature map X_a obtained after adopting an RAC can be formulated by

$$X_a(P) = \text{RAC}(P) = \text{Deformable}(X, \text{offsets}(P)). \quad (5)$$

As shown in Fig. 4(b), the proposed RAC is implemented by a deformable convolution with a kernel size of 3×3. Instead of learning the kernel offsets offsets(P) through back propagation, we compute them as the distances between the sampling points of standard convolution filter and corresponding rotated bounding box at position P [see Fig. 4(a)]. Especially, there is a square anchor at position P and a corresponding rotated bounding box O_p represented as $(x_p, y_p, w_p, h_p, \theta_p)$. However, the center point of O_p is shifted from (x_0, y_0) to (x_p, y_p) with an offset of $o_{\text{ctr}} = (x_p, y_p) - (x_0, y_0)$. For a standard 3×3 convolution filter centered at P, the offsets of nine sampling points relative to P are calculated as $loc_{\text{conv}} = \{(-1, -1), (-1, 0), \ldots, (0, 1), (1, 1)\}$. In order to extract the features $X_a(P)$ of rotated box O_p, we sample nine points on X corresponding to O_p, as shown in Fig. 4(a). Their offsets relative to (x_p, y_p) are loc_{box}, which denotes the distances between the coordinates of nine sampling points and the center point (x_p, y_p) of O_p. Therefore, the offsets between the sampling points of O_p and standard convolution filter can be formulated as

$$\text{offsets}(P) = \{\text{loc}_{\text{box}}(k) - \text{loc}_{\text{conv}}(k) + o_{\text{ctr}}\}, \quad \text{for } k \in \{1, 2, \ldots, 9\} \quad (6)$$

where k denotes the index of sampling points in convolution filter.

The architecture of CRM is shown in Fig. 2. The detection head is identical to the one in BD, and we insert an extra RAC layer in CRM between the backbone network and the detection head, which can accurately extract the features of the predicted rotated box.

During training, unlike BD, CRM refines the rotated bounding box predicted in the previous stage, so the anchor matching is between the rotated box and GT-R objects. Thus, we compute the skew IoU that is calculated between the rotated boxes. Fig. 5 shows the examples of standard IoU between the horizontal boxes and skew IoU between the rotated boxes. We treat the rotated box with IoU higher than the threshold of 0.5 as positive samples, while those with IoU lower than 0.4 are negative samples. The rest are ignored. The formulation of regression targets is the same as BD. The predicted rotated boxes are closer to the ground truth than the anchors in BD; therefore, it can provide a better initialization for regression. This cascade module can be repeated multiple times. The overall loss for the rotated anchor-aided detection module is defined as

$$L_{\text{total}} = \sum_{j=1}^{I} \alpha_j \cdot L_j \quad (7)$$

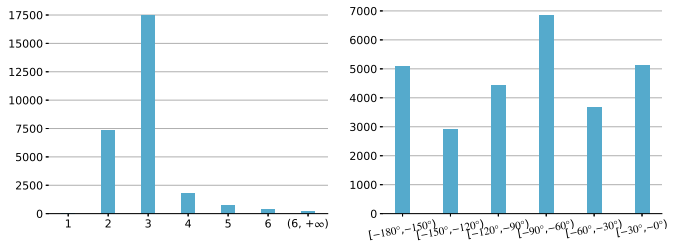
where j and I denotes the index and total num of CRMs, respectively. L_j indicates the loss of the jth stage, and α_j is a hyperparameter that controls the tradeoff between the loss terms of different stages and is set to 1 by default in our experiments.

IV. Experiment

A. Data Set and Evaluation Metric

We evaluate the proposed CR2A-Net on the challenging public data sets DOTA [24] and HRSC2016 [25]. DOTA is a large-scale data set for object detection in remote sensing

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(a) (b)

Fig. 6. Some statistical analysis of the DOTA data set. (a) Quantity of instances with different aspect ratios in DOTA. (b) Quantity of instances with different rotation angles in DOTA.

images. It includes 1411 images for training, 458 images for validation, and 937 images for testing. The images are generated from different sensors and platforms with sizes range from 800 × 800 to 4000 × 4000 pixels. The object instances in images are annotated using quadrilaterals and vary in scales, aspect ratios, and orientations. Fig. 6 analyzes the distribution of ships in the training images. It can be seen that the ships in the DOTA training set are in different orientations and aspect ratios. As labels of images in the testing set are unavailable, we report the detection performance on the testing set by submitting the result files to the public server. The HRSC2016 data set is a high-resolution data set for ship detection that contains images collected from six harbors. The image sizes in the data set range from 300 × 300 to 1500 × 900, and the ship instances in images are annotated using rotated rectangles with various orientations. The data set includes 436 images for training, 181 images for validation, and 444 images for testing, respectively. Because the number of training images is small, we use all the images in the training and validation sets for training.

We adopt average precision (AP) as the evaluation metric in our experiments, which is defined as the same as the PASCAL VOC 2012 object detection challenge [53]. In addition, we will make our code available at https://github.com/yingyu13/CR2A-Net.

B. Implementation Details

1) Data Set Setting: As optical remote sensing images often have a big size, we divide the images into small patches to fit the hardware memory. Especially, the images in the DOTA training set are split into small crops of size 600 × 600 pixels, with an overlap of 150 pixels among neighboring patches. Finally, we obtain 49 152 patches in total. Among these patches, only 3764 ones contain ships. Therefore, we use all the patches to train our classifier network in DPM, while we only use the patches containing ships to train the detector in a rotated anchor-aided detection module. In the detector, images are resized to 800 × 800 to improve the detection of small objects. Random horizontal flipping and random rotation are used for data augmentation. The data cropping operation during the inference stage is the same as the training stage. Especially, we crop the images in the validation set and testing set, respectively, and merge the detection results of small patches to obtain the final detection result of the original images. For the HRSC2016 data set, since most of the image sizes are not very large, therefore, we skip the DPM and resize the images to 800 × 800 directly for the training and testing of rotated anchor-aided detection module. Random horizontal flipping is used for data augmentation.

2) Training Details: We employ the PyTorch [54] framework to implement our method. All the experiments and running efficiency are evaluated on a single TITAN X (Pascal) with 12-GB memory. In all experiments, we adopt the stochastic gradient descent (SGD) optimizer with momentum and weight decay set to 0.0001 and 0.9, respectively. For the classifier, the network is initialized as described in Section III. The batch size is set to 128. The learning rate is set to 0.001 and then divided by 10 every 10 epochs. We train the classifier for 20 epochs in total. For the detector, the batch size is set to 2. The learning rate is initialized with 0.00125 and divided by 10 at every learning rate decay. For the experiments on DOTA, we adopt ResNet50 [49] pretrained on ImageNet [50] with FPN structure as backbone network for feature extraction. We train the detector for a total number of iterations of 50k, and the learning rate decays at 34k and 45k, respectively. For the experiments on HRSC2016, we adopt ResNet101 with FPN as a backbone network for a fair comparison with existing methods. The detector is trained for a total number of iterations of 12k, and the learning rate decays at 8k and 11k, respectively.

TABLE II
EXPERIMENTAL RESULTS OF EACH COMPONENT ON THE DOTA VALIDATION SET

BD	CRM	DPM	mAP(%)	Runtime(s)
			78.83	0.05
√			67.61	0.04
√	√		84.13	0.08
√		√	72.92	0.03
√	√	√	87.22	0.07

C. Ablation Study

To verify the effectiveness of each component of our proposed CR2A-Net, we perform an ablation study on the DOTA validation set. We train several models with different settings for the ablation study. Especially, RetinaNet with a rotational head to predict rotated box (i.e., RetinaNet-R) is employed as a one-stage anchor-based comparison method. The basic rotated anchor-aided detector with ATSS sampling strategy (i.e., BD) is adopted as a baseline network. The BD with DPM included (i.e., BD + DPM) performs detection on the outputs from DPM. The BD with CRM included (i.e., BD + CRM) refines the predicted rotated box from BD, which constitutes our rotated anchor-aided detection module. Some qualitative results are summarized in Table II, and the first line corresponds to the result of RetinaNet-R.

1) Effectiveness of Data Preprocessing Module: As shown in Table II, the baseline network BD achieves an AP of 67.61%. When including the proposed DPM, the performance of BD + DPM can be improved by 5.31% compared with BD. This demonstrates that the DPM is beneficial to improving the detection performance. Meanwhile, we can see that, after adding DPM, the inference speed is increased. The reason is that the DPM can filter out those blank patches without targets and reduce large amounts of false detections.

2) Effectiveness of Rotated Anchor-Aided Detection Module: When including CRM, the performance of BD+CRM can be significantly improved by 16.52%, compared with BD.

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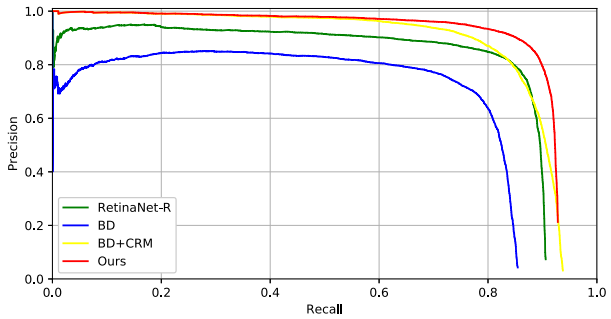


Fig. 7. Precision recall curves of different methods.

TABLE III
EVALUATION WITH DIFFERENT CONVOLUTIONS IN CRM

Methods	Standard Conv	Deformable Conv	Rotated Align Conv	mAP(%)
BD				67.61
BD+CRM	√			65.43
BD+CRM		√		74.81
BD+CRM			√	84.13

This demonstrates that CRM can gradually refine the predicted rotated box and greatly improve the localization accuracy. By employing this coarse-to-fine structure, our rotated anchor-aided detection module can achieve high-precision performance for detecting rotated ships. In addition, BD + CRM outperforms RetinaNet-R by 5.3%, indicating that the proposed anchor-aided network can achieve comparable performance with anchor-based methods, while enjoying a more flexible structure.

Finally, by including all the components, our CR2A-Net outperforms RetinaNet-R by 8.39%, pushing the AP to 87.22%. We can see that the proposed components are effective for improving network performance, and they are complementary to each other. Besides, we plot the P-R curves of different methods in Fig. 7. It can be observed that BD has a lower precision and recall rate than RetinaNet-R. After adding the CRM module, both the precision and recall are improved and surpass RetinaNet-R, indicating that CRM can effectively promote the localization accuracy. The main function of DPM is to reduce false positives, so, after adding the DPM module, our method obtains a noticeable precision improvement at the high recall region where false positives mainly exist.

3) Effectiveness of Rotated Align Convolution: To demonstrate the effectiveness of the proposed RAC, we compare different convolution layers for feature extraction from rotated regions. Especially, we replace the RAC in CRM with a standard convolution layer and a deformable convolution layer, respectively. Qualitative comparisons of the detection results are reported in Table III.

We can observe that model with our proposed RAC gets the best accuracy. Especially, compared with BD, the performance of BD + CRM with standard convolution can be even worse. As we described in Section III, the center point of the rotated box predicted by BD is shifted from the point of the feature map. Meanwhile, the shapes and orientations of predicted boxes are different. Therefore, adopting a standard convolution directly cannot extract robust rotation-invariance features from the corresponding rotated regions. The AP score of BD + CRM with deformable convolution is better than BD. After employing RAC, we can further promote the performance by 9.32% over deformable convolution. This demonstrates that the proposed RAC is superior in extracting features from rotated regions accurately and plays a crucial role in CRM.

TABLE IV
IMPACT OF ITERATIVE REFINEMENT

Num of Cascade Refinement Modules	mAP(%)
Baseline	67.61
I=1	84.13
I=2	73.38
I=3	72.42
I=4	64.65

TABLE V
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA TESTING SET

Methods	Backbone	mAP(%)
FR-O [24]	ResNet101	36.2
R2CNN [28]	VGG16	55.81
CAD-Net [45]	ResNet101-FPN	76.7
APE [55]	ResNet50-FPN	79.54
RoI-Transformer [44]	ResNet101-FPN	83.59
RetinaNet [13]	ResNet50-FPN	75.24
IENet [56]	ResNet101-FPN	65.01
Axis Learning [57]	ResNet101-FPN	76.3
O2-DNet [58]	104-Hourglass	78.62
Ours	ResNet50-FPN	84.12

In addition, we perform an experiment to explore the impact of iterative refinement, i.e., the num of CRMs I. The experimental results are shown in Table IV. It can be observed that adding multiple refinement modules does not necessarily bring continuous improvement, and the model gets the best accuracy when adding one CRM.

D. Comparison With State-of-the-Art Methods

We compare the proposed CR2A-Net with several state-of-the-art methods on the DOTA and HRSC2016 data sets. Our method outperforms all other comparison methods on these data sets.

1) DOTA Data Set: For the experiments on DOTA, the comparison methods can be divided into two categories, i.e., anchor-based methods, such as FR-O [24], R2CNN [28], CAD-Net [45], APE [55], RoI-Transformer [44], and RetinaNet [13], and anchor-free methods, such as IENet [56], Axis Learning [57], and O2-DNet [58]. The results on the DOTA testing set are shown in Table V. Note that some of these methods adopt different backbone networks from our method. The detection results of these methods on the DOTA testing set are taken from the original articles. As shown in Table V, anchor-based RoI-Transformer achieves leading performance among all comparison methods. Our method obtains an improvement on RoI-Transformer and achieves the best performance, demonstrating its effectiveness for ship detection in remote sensing images. In addition, our method outperforms other anchor-free methods by a large margin up to 5.5%, benefitting from the complementary effect of the proposed components. Besides, we indicate that employing a more powerful backbone network and tricks, such as multiscale training and testing, will further boost the detection performance.

As the labels of images in the testing set are unavailable, we further implement some baseline methods whose source

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TABLE VI
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA VALIDATION SET

Methods	Backbone	mAP(%)	Runtime(s)
YOLO-v3-R	Darknet53	72.87	0.02
CenterNet-R	ResNet101	76.06	0.11
RetinaNet-R	ResNet50-FPN	78.83	0.05
Faster R-CNN [10]	ResNet50-FPN	82.89	0.05
RoI-Transformer [44]	ResNet50-FPN	86.26	0.07
Mask R-CNN [15]	ResNet50-FPN	82.71	0.05
Cascade Mask R-CNN	ResNet50-FPN	84.69	0.09
Hybrid Task Cascade [59]	ResNet50-FPN	84.56	0.09
Ours	ResNet50-FPN	87.22	0.07
	ResNet101-FPN	86.62	0.13
	ResNet152-FPN	86.91	0.16

TABLE VII
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE HRSC2016 DATA SET

Methods	Backbone	mAP(%)	Runtime(s)
YOLO-v3-R	Darknet53	79.53	0.03
CenterNet-R	ResNet101	83.72	0.12
RetinaNet-R	ResNet101-FPN	82.89	0.07
RRD [60]	VGG16	84.3	slow
R2CNN [28]	ResNet101-FPN	73.07	0.5
RC2 [61]	VGG16	75.7	>1
RRPN [29]	ResNet101-FPN	79.08	0.28
RoI-Transformer [44]	ResNet101-FPN	86.2	0.11
Ours	ResNet101-FPN	90.79	0.10
	ResNet152-FPN	91.05	0.12
	ResNeXt101-FPN	90.57	0.16

codes are released on the DOTA validation set so that the running efficiency of each method can be evaluated fairly on the same hardware, i.e., a single TITAN X (Pascal) with 12-GB memory. The comparative results of speed and accuracy are reported in Table VI. Note that the runtime per image includes the postprocessing of NMS. These methods can be divided into regression-based methods, such as YOLO-v3-R, CenterNet-R, RetinaNet-R, Faster R-CNN [10], and RoI-Transformer [44], and segmentation-based methods, such as Mask R-CNN [15], Cascade Mask R-CNN, and Hybrid Task Cascade [59]. The YOLO-v3-R and CenterNet-R algorithms are implemented based on general YOLO-v3 [62] and CenterNet [17] designed for horizontal object detection, thus enabling the models to produce rotated boxes. Because the segmentation masks can be converted to rotated boxes directly, segmentation-based methods can be used to perform rotated object detection as well. From Table VI, we can observe that our method outperforms all these comparison methods. Especially, our CR2A-Net obtains an mAP of 87.22%, which is significantly higher than the baseline methods, achieving improvements of 0.96%, 2.53%, 2.66%, 4.33%, 4.51%, 8.39%, 14.35%, and 11.16% over RoI-Transformer, Cascade Mask R-CNN, Hybrid Task Cascade, Faster R-CNN, Mask R-CNN, RetinaNet-R, YOLO-v3-R, and CenterNet-R. This indicates that our CR2A-Net performs very well in detecting arbitrary-oriented ships.

2) HRSC2016 Data Set: For the experiments on HRSC2016, we compare our CR2A-Net with the existing methods, such as YOLO-v3-R, CenterNet-R, RetinaNet-R, RRD [60], R2CNN [28], RC2 [61], RRPN [29], and RoI-Transformer [44]. The comparison results are summarized in Table VII. It can be seen that the proposed CR2A-Net obtains an mAP of 90.79% and achieves the best performance. Especially, our CR2A-Net significantly outperforms the baseline methods, achieving improvements of 4.59%, 11.71%, 15.09%, 17.72%, 6.49%, 7.9%, 11.26%, and 7.07% over RoI-Transformer, RRPN, RC2, R2CNN, RRD, RetinaNet-R, YOLO-v3-R, and CenterNet-R, respectively. This illustrates the effectiveness of our CR2A-Net in detecting arbitrary-oriented ships with large aspect ratios in remote sensing images. Moreover, the above experimental results on these two data sets further demonstrate the robustness of the proposed method when adapting to different data sets with different object sizes and orientations. The reason is that we adopt an anchor-aided strategy that avoids the complicated anchor related settings and bring better robustness to the model.

In addition, we conduct experiments to compare the performance of the proposed model with different backbones. The experimental results are shown in Tables VI and VII. We replace the ResNet50-FPN backbone network on the DOTA data set with ResNet101-FPN and ResNet152-FPN, respectively. Meanwhile, we replace the ResNet101-FPN backbone network on the HRSC2016 data set with ResNet152-FPN and ResNeXt101-FPN [63]. As can be observed, the performance is decreased slightly when using a larger backbone network. This indicates that adopting a more advanced backbone network may not necessarily bring performance improvements but will increase the inference time inevitably. Therefore, we choose the backbone in this article considering the tradeoff of efficiency and accuracy, as well as making a fair comparison with the previous methods. Besides, as can be seen from Tables VI and VII, our CR2A-Net can achieve impressively better detection performance with a comparable speed.

E. Visual Results

To demonstrate the advantages of our method over previous ones, we show some visual comparison results. Fig. 8 shows the detection results of BD and BD + CRM with and without the data preprocessing module, respectively. It can be seen that both BD and BD + CRM produce a lot of false positives for the image without targets. In this image, BD + CRM has more false positives than BD. This is because BD + CRM will slightly increase the false positives while greatly improving the recall rate. Fortunately, they usually have a relatively low classification score and can be filtered out. As can be seen from the figure, after adding the DPM, these false positives are eliminated, indicating that the data preprocessing module can accurately distinguish images without targets and effectively filter the false detections on these empty images.

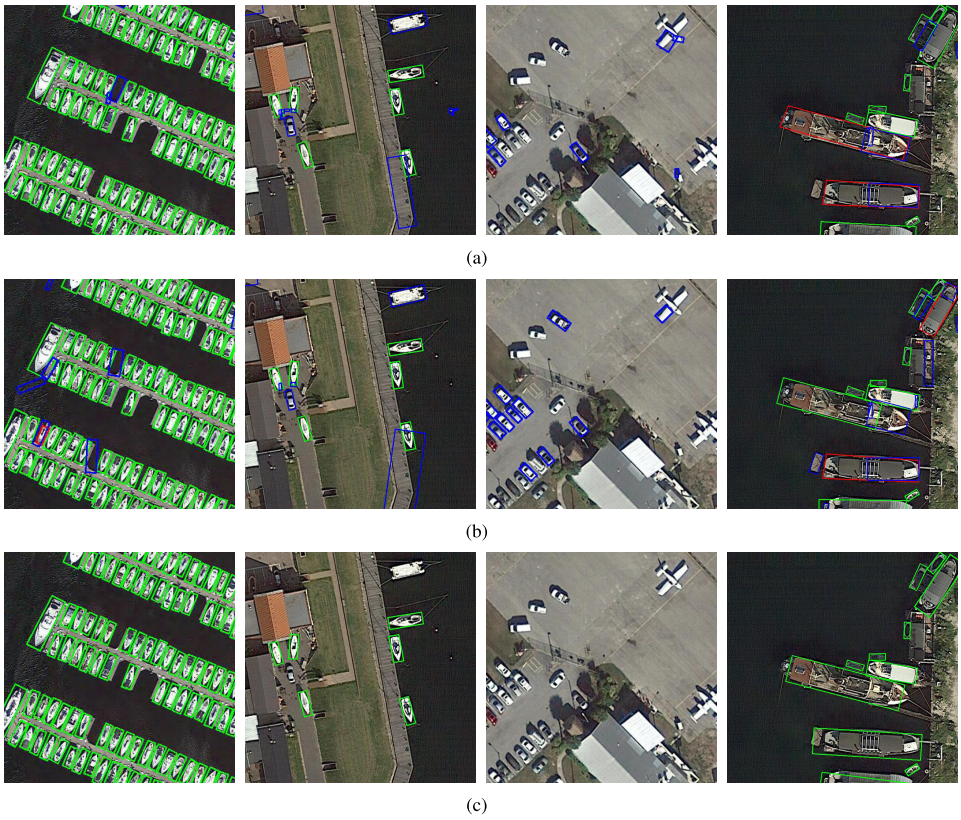
Fig. 9 shows some detection results of different methods on sample images from the DOTA data set in various scenarios. The first row is RetinaNet-R, the second row is Faster R-CNN, and the third row is our method. It can be seen that RetinaNet-R and Faster R-CNN often produce false detections in different situations. For example, in the first and second images where the ships are densely and sparsely distributed, respectively, some false positives and false negatives occur. In the third image without ship targets, some planes are wrongly detected as ships. In the fourth image that contains objects of different sizes, the ships with a large aspect ratio are missed. As a comparison, our CR2A-Net can detect all the ships correctly in different scenarios and produce much fewer false positives. It indicates that our method is robust against the impact of different scenarios and can achieve an

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(a) (b)

Fig. 8. Visualization of detection results before and after using DPM. The blue boxes denote false positives. (a) Detection results of the BD. (b) Detection results of the BD + CRM.



(a) (b) (c)

Fig. 9. Comparison of the detection results by different methods. The green, blue, and red boxes indicate the detected ships, false positives, and false negatives. (a) RetinaNet-R. (b) Faster R-CNN. (c) Our proposed CR2A-Net.

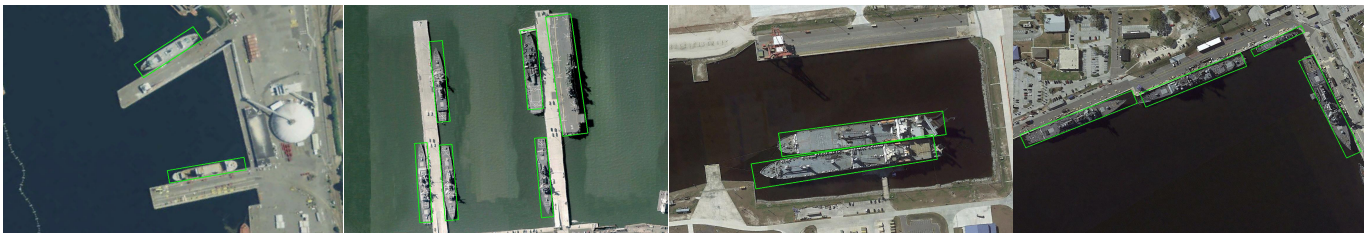


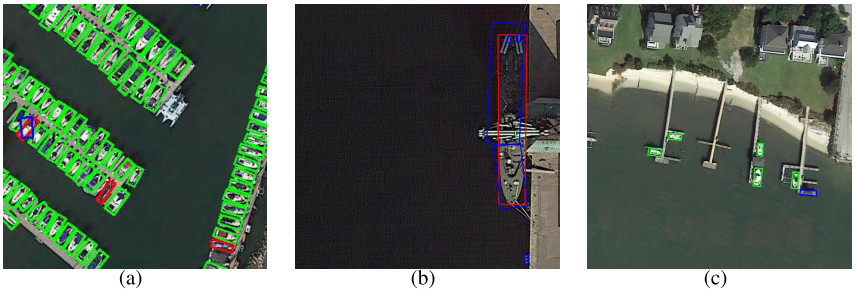
Fig. 10. Some detection results of the proposed CR2A-Net on the HRSC2016 data set.

outstanding detection performance. This is mainly attributed to the combination of our DPM, anchor-aided detector, and CRM. In addition, we visualize some detection results of our method on the HRSC2016 data set in Fig. 10. As can be seen from Fig. 10, the proposed method can output pleasing results and accurately detect arbitrary-oriented ships even with large aspect ratios.

F. Analysis of Failure Cases

We show some undesirable detection results in Fig. 11. In Fig. 11(a), some small ships are missed in the case of dense distribution. Although there is a ship target, the blue box is judged to be a false positive, which is because the angle prediction is not accurate enough. In Fig. 11(b), the ship with a large aspect ratio is missed, and some false positives are produced.

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(a) (b) (c)

Fig. 11. Examples of some failure cases on DOTA. (a) Dense objects. (b) Single object. (c) Small objects. The green boxes, blue boxes, and red boxes indicate the detected ships, false positives, and false negatives.

This phenomenon is largely due to the lack of corresponding training samples in the training set. In Fig. 11(c), the portland is wrongly detected as a ship because they share a similar appearance. We believe that these problems can be solved by introducing a more accurate angle prediction, more abundant context information, and a more extensive data set, which will be further researched in our future work.

V. Conclusion

In this article, we propose an accurate and powerful CR2A-Net for ship detection in high-resolution remote sensing images. We develop a data preprocessing embedded cascade structure to filter the images without targets before detecting. The rotated anchor-aided detector is designed in a coarse-to-fine architecture to achieve high-precision performance for rotated ships. Meanwhile, the anchor-aided design enables the network to perform detection in a more flexible manner and break through the limitations of anchor-based methods. Furthermore, an RAC is proposed to better extract features from rotated regions and greatly improve the detection accuracy. Extensive experiments on the challenging DOTA and HRSC2016 data sets show that the proposed CR2A-Net can achieve state-of-the-art ship detection performance.

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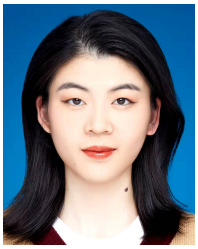
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